1.

```
--- MEMORY ALLOCATION TEST ---

Before allocation:

========== MEMORY STATISTICS ==========
Total allocated pages : 425
Total freed pages     : 214
Currently allocated   : 211
Available pages       : 56789
=======================================

Allocating 3 pages...

After allocation:

========== MEMORY STATISTICS ==========
Total allocated pages : 428
Total freed pages     : 214
Currently allocated   : 214
Available pages       : 56786
=======================================

Releasing 3 pages...

After deallocation:

========== MEMORY STATISTICS ==========
Total allocated pages : 428
Total freed pages     : 217
Currently allocated   : 211
Available pages       : 56789
=======================================
```

2.

```
$ vttest

--- VIRTUAL TO PHYSICAL ADDRESS TEST ---

Translating valid addresses:

Page 0:
Virtual Address  : 3000
Physical Address : dee3000

Page 1:
Virtual Address  : 4000
Physical Address : dee1000

Page 2:
Virtual Address  : 5000
Physical Address : df22000

Testing invalid address:
Invalid or unmapped virtual address: 70000000
$
```

3.

```
$ pagetest

--- PAGE PROTECTION TEST ---

Initial page information:

--- PAGE INFORMATION ---
Virtual Address : 3000
Physical Frame  : df2b
Present (P)     : 1
Writable (W)    : 1
User (U)        : 1

Creating child process for protection test...

Child: Changing page to read-only

--- PAGE INFORMATION ---
Virtual Address : 3000
Physical Frame  : df21
Present (P)     : 1
Writable (W)    : 1
User (U)        : 1
Page at virtual address 3000 is now READ-ONLY
```

```
Page information after protection:

--- PAGE INFORMATION ---
Virtual Address : 3000
Physical Frame  : df21
Present (P)     : 1
Writable (W)    : 0
User (U)        : 1

Child: Attempting to write to read-only page...
pid 6 pagetest: trap 14 err 7 on cpu 0 eip 0xbb addr 0x3000--kic

Parent: Child finished after protection test.

--- TEST COMPLETED ---
$
```

4.

```
====================================
    PAGE REPLACEMENT SIMULATION
====================================


******** FRAME COUNT: 3 ********

========== FIFO (3 FRAMES) ==========
Page 7: FAULT [ 7  -  - ]
Page 0: FAULT [ 7  0  - ]
Page 1: FAULT [ 7  0  1 ]
Page 2: FAULT [ 2  0  1 ]
Page 0: HIT [ 2  0  1 ]
Page 3: FAULT [ 2  3  1 ]
Page 0: FAULT [ 2  3  0 ]
Page 4: FAULT [ 4  3  0 ]
Page 2: FAULT [ 4  2  0 ]
Page 3: FAULT [ 4  2  3 ]
Page 0: FAULT [ 0  2  3 ]
Page 3: HIT [ 0  2  3 ]
Page 2: HIT [ 0  2  3 ]
Page 1: FAULT [ 0  1  3 ]
Page 2: FAULT [ 0  1  2 ]
Page 0: HIT [ 0  1  2 ]
Page 1: HIT [ 0  1  2 ]
Page 7: FAULT [ 7  1  2 ]
Page 0: FAULT [ 7  0  2 ]
Page 1: FAULT [ 7  0  1 ]
Page Faults = 15
Page Hits   = 5
Hit Ratio   = 5/20
Fault Ratio = 15/20
```

```
========== LRU (3 FRAMES) ==========
Page 7: FAULT [ 7  -  - ]
Page 0: FAULT [ 7  0  - ]
Page 1: FAULT [ 7  0  1 ]
Page 2: FAULT [ 2  0  1 ]
Page 0: HIT [ 2  0  1 ]
Page 3: FAULT [ 2  0  3 ]
Page 0: HIT [ 2  0  3 ]
Page 4: FAULT [ 4  0  3 ]
Page 2: FAULT [ 4  0  2 ]
Page 3: FAULT [ 4  3  2 ]
Page 0: FAULT [ 0  3  2 ]
Page 3: HIT [ 0  3  2 ]
Page 2: HIT [ 0  3  2 ]
Page 1: FAULT [ 1  3  2 ]
Page 2: HIT [ 1  3  2 ]
Page 0: FAULT [ 1  0  2 ]
Page 1: HIT [ 1  0  2 ]
Page 7: FAULT [ 1  0  7 ]
Page 0: HIT [ 1  0  7 ]
Page 1: HIT [ 1  0  7 ]
Page Faults = 12
Page Hits   = 8
Hit Ratio   = 8/20
Fault Ratio = 12/20
```

```
========== OPTIMAL (3 FRAMES) ==========
Page 7: FAULT [ 7  -  - ]
Page 0: FAULT [ 7  0  - ]
Page 1: FAULT [ 7  0  1 ]
Page 2: FAULT [ 2  0  1 ]
Page 0: HIT [ 2  0  1 ]
Page 3: FAULT [ 2  0  3 ]
Page 0: HIT [ 2  0  3 ]
Page 4: FAULT [ 2  4  3 ]
Page 2: HIT [ 2  4  3 ]
Page 3: HIT [ 2  4  3 ]
Page 0: FAULT [ 2  0  3 ]
Page 3: HIT [ 2  0  3 ]
Page 2: HIT [ 2  0  3 ]
Page 1: FAULT [ 2  0  1 ]
Page 2: HIT [ 2  0  1 ]
Page 0: HIT [ 2  0  1 ]
Page 1: HIT [ 2  0  1 ]
Page 7: FAULT [ 7  0  1 ]
Page 0: HIT [ 7  0  1 ]
Page 1: HIT [ 7  0  1 ]
Page Faults = 9
Page Hits   = 11
Hit Ratio   = 11/20
Fault Ratio = 9/20
```

```
******** FRAME COUNT: 4 ********

========== FIFO (4 FRAMES) ==========
Page 7: FAULT [ 7  -  -  - ]
Page 0: FAULT [ 7  0  -  - ]
Page 1: FAULT [ 7  0  1  - ]
Page 2: FAULT [ 7  0  1  2 ]
Page 0: HIT [ 7  0  1  2 ]
Page 3: FAULT [ 3  0  1  2 ]
Page 0: HIT [ 3  0  1  2 ]
Page 4: FAULT [ 3  4  1  2 ]
Page 2: HIT [ 3  4  1  2 ]
Page 3: HIT [ 3  4  1  2 ]
Page 0: FAULT [ 3  4  0  2 ]
Page 3: HIT [ 3  4  0  2 ]
Page 2: HIT [ 3  4  0  2 ]
Page 1: FAULT [ 3  4  0  1 ]
Page 2: FAULT [ 2  4  0  1 ]
Page 0: HIT [ 2  4  0  1 ]
Page 1: HIT [ 2  4  0  1 ]
Page 7: FAULT [ 2  7  0  1 ]
Page 0: HIT [ 2  7  0  1 ]
Page 1: HIT [ 2  7  0  1 ]
Page Faults = 10
Page Hits   = 10
Hit Ratio   = 10/20
Fault Ratio = 10/20
```

```
========== LRU (4 FRAMES) ==========
Page 7: FAULT [ 7  -  -  - ]
Page 0: FAULT [ 7  0  -  - ]
Page 1: FAULT [ 7  0  1  - ]
Page 2: FAULT [ 7  0  1  2 ]
Page 0: HIT [ 7  0  1  2 ]
Page 3: FAULT [ 3  0  1  2 ]
Page 0: HIT [ 3  0  1  2 ]
Page 4: FAULT [ 3  0  4  2 ]
Page 2: HIT [ 3  0  4  2 ]
Page 3: HIT [ 3  0  4  2 ]
Page 0: HIT [ 3  0  4  2 ]
Page 3: HIT [ 3  0  4  2 ]
Page 2: HIT [ 3  0  4  2 ]
Page 1: FAULT [ 3  0  1  2 ]
Page 2: HIT [ 3  0  1  2 ]
Page 0: HIT [ 3  0  1  2 ]
Page 1: HIT [ 3  0  1  2 ]
Page 7: FAULT [ 7  0  1  2 ]
Page 0: HIT [ 7  0  1  2 ]
Page 1: HIT [ 7  0  1  2 ]
Page Faults = 8
Page Hits   = 12
Hit Ratio   = 12/20
Fault Ratio = 8/20
```

```
========== OPTIMAL (4 FRAMES) ==========
Page 7: FAULT [ 7  -  -  - ]
Page 0: FAULT [ 7  0  -  - ]
Page 1: FAULT [ 7  0  1  - ]
Page 2: FAULT [ 7  0  1  2 ]
Page 0: HIT [ 7  0  1  2 ]
Page 3: FAULT [ 3  0  1  2 ]
Page 0: HIT [ 3  0  1  2 ]
Page 4: FAULT [ 3  0  4  2 ]
Page 2: HIT [ 3  0  4  2 ]
Page 3: HIT [ 3  0  4  2 ]
Page 0: HIT [ 3  0  4  2 ]
Page 3: HIT [ 3  0  4  2 ]
Page 2: HIT [ 3  0  4  2 ]
Page 1: FAULT [ 1  0  4  2 ]
Page 2: HIT [ 1  0  4  2 ]
Page 0: HIT [ 1  0  4  2 ]
Page 1: HIT [ 1  0  4  2 ]
Page 7: FAULT [ 1  0  7  2 ]
Page 0: HIT [ 1  0  7  2 ]
Page 1: HIT [ 1  0  7  2 ]
Page Faults = 8
Page Hits   = 12
Hit Ratio   = 12/20
Fault Ratio = 8/20
```

```
******** FRAME COUNT: 5 ********

========== FIFO (5 FRAMES) ==========
Page 7: FAULT [ 7  -  -  -  - ]
Page 0: FAULT [ 7  0  -  -  - ]
Page 1: FAULT [ 7  0  1  -  - ]
Page 2: FAULT [ 7  0  1  2  - ]
Page 0: HIT [ 7  0  1  2  - ]
Page 3: FAULT [ 7  0  1  2  3 ]
Page 0: HIT [ 7  0  1  2  3 ]
Page 4: FAULT [ 4  0  1  2  3 ]
Page 2: HIT [ 4  0  1  2  3 ]
Page 3: HIT [ 4  0  1  2  3 ]
Page 0: HIT [ 4  0  1  2  3 ]
Page 3: HIT [ 4  0  1  2  3 ]
Page 2: HIT [ 4  0  1  2  3 ]
Page 1: HIT [ 4  0  1  2  3 ]
Page 2: HIT [ 4  0  1  2  3 ]
Page 0: HIT [ 4  0  1  2  3 ]
Page 1: HIT [ 4  0  1  2  3 ]
Page 7: FAULT [ 4  7  1  2  3 ]
Page 0: FAULT [ 4  7  0  2  3 ]
Page 1: FAULT [ 4  7  0  1  3 ]
Page Faults = 9
Page Hits   = 11
Hit Ratio   = 11/20
Fault Ratio = 9/20
```

```
========== LRU (5 FRAMES) ==========
Page 7: FAULT [ 7  -  -  -  - ]
Page 0: FAULT [ 7  0  -  -  - ]
Page 1: FAULT [ 7  0  1  -  - ]
Page 2: FAULT [ 7  0  1  2  - ]
Page 0: HIT [ 7  0  1  2  - ]
Page 3: FAULT [ 7  0  1  2  3 ]
Page 0: HIT [ 7  0  1  2  3 ]
Page 4: FAULT [ 4  0  1  2  3 ]
Page 2: HIT [ 4  0  1  2  3 ]
Page 3: HIT [ 4  0  1  2  3 ]
Page 0: HIT [ 4  0  1  2  3 ]
Page 3: HIT [ 4  0  1  2  3 ]
Page 2: HIT [ 4  0  1  2  3 ]
Page 1: HIT [ 4  0  1  2  3 ]
Page 2: HIT [ 4  0  1  2  3 ]
Page 0: HIT [ 4  0  1  2  3 ]
Page 1: HIT [ 4  0  1  2  3 ]
Page 7: FAULT [ 7  0  1  2  3 ]
Page 0: HIT [ 7  0  1  2  3 ]
Page 1: HIT [ 7  0  1  2  3 ]
Page Faults = 7
Page Hits   = 13
Hit Ratio   = 13/20
Fault Ratio = 7/20
```

```
========== OPTIMAL (5 FRAMES) ==========
Page 7: FAULT [ 7  -  -  -  - ]
Page 0: FAULT [ 7  0  -  -  - ]
Page 1: FAULT [ 7  0  1  -  - ]
Page 2: FAULT [ 7  0  1  2  - ]
Page 0: HIT [ 7  0  1  2  - ]
Page 3: FAULT [ 7  0  1  2  3 ]
Page 0: HIT [ 7  0  1  2  3 ]
Page 4: FAULT [ 4  0  1  2  3 ]
Page 2: HIT [ 4  0  1  2  3 ]
Page 3: HIT [ 4  0  1  2  3 ]
Page 0: HIT [ 4  0  1  2  3 ]
Page 3: HIT [ 4  0  1  2  3 ]
Page 2: HIT [ 4  0  1  2  3 ]
Page 1: HIT [ 4  0  1  2  3 ]
Page 2: HIT [ 4  0  1  2  3 ]
Page 0: HIT [ 4  0  1  2  3 ]
Page 1: HIT [ 4  0  1  2  3 ]
Page 7: FAULT [ 7  0  1  2  3 ]
Page 0: HIT [ 7  0  1  2  3 ]
Page 1: HIT [ 7  0  1  2  3 ]
Page Faults = 7
Page Hits   = 13
Hit Ratio   = 13/20
Fault Ratio = 7/20


--- ALL ALGORITHMS COMPLETED ---
$
```